CPSC 304 Project Cover Page

Milestone #: ___2____

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Group Number: ___92___

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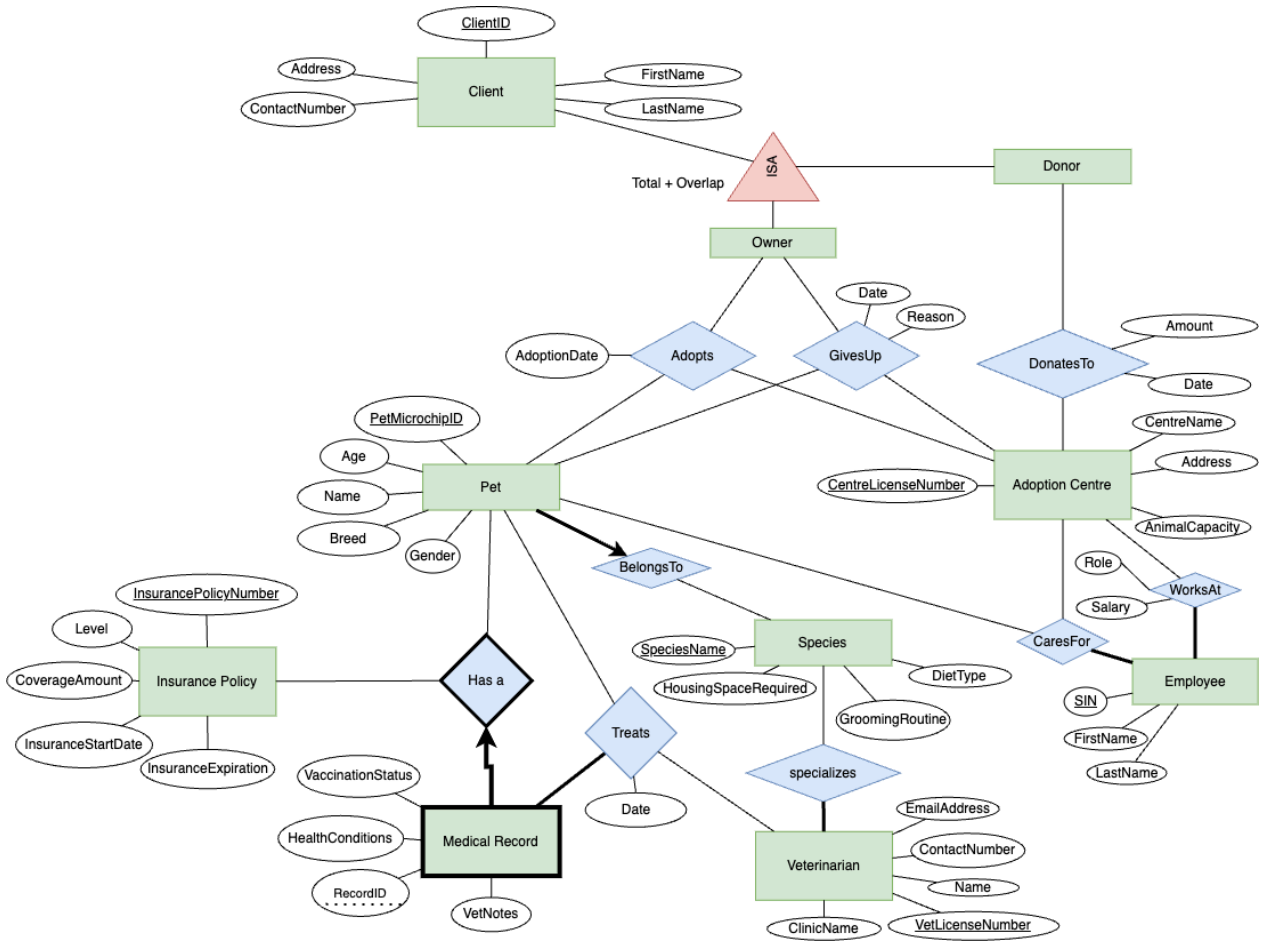
By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

2. A brief (~2-3 sentences) summary of your project. Many of your TAs are managing multiple projects so this will help them remember details about your project.

Our project focuses on keeping track of pets and their care and management. This is essentially the creation of a centralized animal adoption centre database and medical/care history to ensure their proper well-being.

3. The ER diagram you are basing your item #3 (below) on. This ER diagram may be the same as your milestone 1 submission or it might be different. If you have made changes from the version submitted in milestone 1, attach a note indicating what changes have been made and why.



In our ER diagram, we changed the following from the original:

- We removed the Houses relationship between Pet and Adoption Centre as it was easier to keep track of the pet's location through the Adopts and GivesUp relations without houses also being present
- We removed the CoveredBy relation between MedicalRecord and InsurancePolicy, as it was redundant with the HasA relationship.
- We added the Role attribute to Employee

- We changed the Adopt and GiveUp relationships to be many-to-many
- We removed Diagnoses between MedicalRecord and Veterinarian, and instead made it a ternary relationship Treats with Pet
- We changed the CaresFor relationship to simply be binary between pet and employee, and not ternary with the adoption centre

4. The schema derived from your ER diagram (above). For the translation of the ER diagram to the relational model, follow the same instructions as in your lectures. The process should be reasonably straightforward. For each table:
    a. List the table definition (e.g., Table1(attr1: domain1, attr2: domain2, ...)). Make sure to include the domains for each attribute.

- Entity Schema -

Client(ClientID: INTEGER(10), FirstName: VARCHAR NOT NULL, LastName: VARCHAR, Address: VARCHAR, ContactNumber: INTEGER)

Owner(**ClientID:** INTEGER(10))
Donor(**ClientID:** INTEGER(10))

Pet(PetMicrochipID: INTEGER(15), Name: VARCHAR, Age: INTEGER, Breed VARCHAR NOT NULL, Gender: VARCHAR, **SpeciesName**: VARCHAR NOT NULL)

InsurancePolicy(InsurancePolicyNumber: INTEGER(20), Level: VARCHAR, CoverageAmount: INTEGER NOT NULL, InsuranceStartDate: CHAR(8), InsuranceExpiration: CHAR(8))

MedicalRecord(**PetMicrochipID**: INTEGER(15), RecordID: INTEGER(10), **InsurancePolicyNumber**: INTEGER(20), VaccinationStatus: CHAR(1), HealthCondition: VARCHAR, VetNotes: VARCHAR)

AdoptionCenter(CenterLicenseNumber: INTEGER(10), CenterName: VARCHAR, Address: VARCHAR, AnimalCapacity: INTEGER)

Species(SpeciesName: VARCHAR, HousingSpaceRequired: VARCHAR, GroomingRoutine: VARCHAR, DietType: VARCHAR)

Veterinarian(VetLicenseNumber: INTEGER(10), Name: VARCHAR NOT NULL, ClinicName: VARCHAR, ContactNumber: INTEGER, EmailAddress: VARCHAR)

Employee(SIN: INTEGER(9), FirstName: VARCHAR, LastName: VARCHAR)

- Relational Schema -

OwnerAdoptsPetFromAdoptionCenter(**PetMicrochipID:** INTEGER(15), AdoptionDate: CHAR(8), **ClientID**: INTEGER(10), **CenterLicenseNumber**: INTEGER(10))

OwnerGivesUpPetToAdoptionCenter(**PetMicrochipID:** INTEGER(15), **ClientID**: INTEGER(10), Date: CHAR(8), Reason: VARCHAR, **CenterLicenseNumber**: INTEGER(10))

DonorDonatesToAdoptionCenter(**ClientID:** INTEGER(10), **CenterLicenseNumber:** INTEGER(10), Amount: INTEGER, Date: CHAR(8))

EmployeeCaresForPet(**PetMicrochipID:** INTEGER(15), **SIN**: INTEGER(9))

EmployeeWorksAtAdoptionCenter(**SIN:** INTEGER(9), **CenterLicenseNumber:** INTEGER(10), Role: VARCHAR, Salary: INTEGER)

VeterinarianSpecializesInSpecies(**VetLicenseNumber:** INTEGER(10), **SpeciesName**: VARCHAR)

VeterinarianTreatsMedicalRecord(**PetMicrochipID**: INTEGER(15), **RecordID:** INTEGER(20), **VetLicenseNumber:** INTEGER(10), Date: CHAR(8))

PetHasAMedicalRecordAndInsurancePolicy(**PetMicrochipID**: INTEGER(15), **InsurancePolicyNumber:** INTEGER(20), **RecordID:** INTEGER(20))

b. Specify the primary key (PK), candidate key, (CK) foreign keys (FK), and other constraints (e.g., not null, unique, etc.) that the table must maintain.

5. Functional Dependencies (FDs)

a. Identify the functional dependencies in your relations, including the ones involving all candidate keys (including the primary key).

PKs and CKs are considered functional dependencies and should be included in the list of FDs. You do not need to include trivial FDs such as A -> A.
Note: In your list of FDs, there must be some kind of valid FD other than those identified by a PK or CK. If you observe that no relations have FDs other than the PK and CK(s), then you will have to intentionally add some (meaningful)

attributes to show valid FDs. We want you to get a good normalization exercise. Your design must go through a normalization process. You do not need to have a non-PK/CK FD for each relation but be reasonable. If your TA feels that some non-PK/CK FDs have been omitted, your grade will be adjusted accordingly.

Client:
ClientID -> FirstName,LastName,Address,ContactNumber

Owner:
ClientID -> FirstName,LastName,Address,ContactNumber

Donor:
ClientID -> FirstName,LastName,Address,ContactNumber

Pet:
PetMicrochipID -> Name, Age, Breed, Gender, SpeciesName
Breed -> SpeciesName

InsurancePolicy:
InsurancePolicyNumber -> Level, CoverageAmount, InsuranceStartDate, InsuranceExpiration
InsuranceStartDate -> InsuranceExpiration

MedicalRecord:
PetMicrochipID,RecordID ->VaccinationStatus,InsurancePolicyNumber,HealthCondition,VetNotes
PetMicrochipID -> VaccinationStatus

AdoptionCenter:
CenterLicenseNumber -> CenterName, Address, AnimalCapacity

Species:
SpeciesName -> HousingSpaceRequired, GroomingRoutine, DietType

Veterinarian:
VetLicenseNumber -> Name,ClinicName,ContactNumber,EmailAddress

Employee:
SIN -> FirstName, LastName

OwnerAdoptsPetFromAdoptionCenter:
PetMicrochipID, AdoptionDate-> ClientID
PetMicrochipID, ClientID -> AdoptionDate

OwnerGivesUpPetToAdoptionCenter:
PetMicrochipID, Date -> Reason,CenterLicenseNumber, ClientID
PetMicrochipID, ClientID -> Date, Reason, CenterLicenseNumber
PetMicrochipID, CenterLicenseNumber -> Date, ClientID, Reason

DonorDonatesToAdoptionCenter:
ClientID,CenterLicenseNumber, Date -> Amount

EmployeeCaresForPet(**PetMicrochipID:** INTEGER(15), **SIN:** INTEGER(9))

- No non-trivial dependencies

EmployeeWorksAtAdoptionCenter:
SIN, CenterLicenseNumber -> Role, Salary
CenterLicenseNumber, Role -> Salary

VeterinarianSpecializesInSpecies(**VetLicenseNumber:** INTEGER(10), **SpeciesName**: VARCHAR)

- No non-trivial dependencies

VeterinarianTreatsMedicalRecord:
PetMicrochipID, RecordID, VetLicenseNumber -> Date

PetHasAMedicalRecordAndInsurancePolicy:

- No non-trivial dependencies

6. Normalization
   a. Normalize each of your tables to be in 3NF or BCNF. Give the list of tables, their primary keys, their candidate keys, and their foreign keys after normalization.

   You should show the steps taken for the decomposition in a manner similar to that done in class. Should there be errors, and no work is shown, no partial credit can be awarded without steps shown.
   The format should be the same as Step 3, with tables listed similar to Table1(attr1:domain1, attr2:domain2, …). ALL Tables must be listed, not only the ones post normalization.

Client:
ClientID -> FirstName,LastName,Address,ContactNumber

This tables is already in BCNF, thus remains:
Client(ClientID: INTEGER(10), FirstName: VARCHAR NOT NULL, LastName: VARCHAR, Address: VARCHAR NOT NULL, ContactNumber: INTEGER)

Pet:
PetMicrochipID -> Name, Age, Breed, Gender, SpeciesName
Breed -> SpeciesName

Breed+ = {Breed, SpeciesName} Which is not in BCNF, thus we must decompose into

BreedSpecies(Breed: VARCHAR, SpeciesName VARCHAR NOT NULL)
Pet(PetMicrochipID: INTEGER(15), Name: VARCHAR, Age: INTEGER, **Breed** VARCHAR NOT NULL, Gender: VARCHAR)

Owner:
ClientID -> FirstName,LastName,Address,ContactNumber

This tables is already in BCNF, thus remains:
Owner(**ClientID:** INTEGER(10))

Donor:
ClientID -> FirstName,LastName,Address,ContactNumber

This tables is already in BCNF, thus remains:
Donor(**ClientID:** INTEGER(10))

InsurancePolicy:
InsurancePolicyNumber -> Level, CoverageAmount, InsuranceStartDate, InsuranceExpiration
InsuranceStartDate -> InsuranceExpiration

InsuranceStartDate+ = {InsuranceStartDate, InsuranceExpiration} Which is not a superkey thus not in BCNF, thus we decompose.

InsuranceDates(InsuranceStartDate, InsuranceExpiration)
InsurancePolicyDecomposed(InsurancePolicyNumber, Level, CoverageAmount, InsuranceStartDate)

These are both in BCNF, thus they are our final tables.
InsuranceDates(InsuranceStartDate : CHAR(8), InsuranceExpiration: CHAR(8))
InsurancePolicyDecomposed(InsurancePolicyNumber: INTEGER(20), Level: VARCHAR, CoverageAmount: INTEGER NOT NULL, **InsuranceStartDate**: CHAR(8))

MedicalRecord:
PetMicrochipID,RecordID ->VaccinationStatus,InsurancePolicyNumber,HealthCondition,VetNotes
PetMicrochipID -> VaccinationStatus

PetMicrochipID+ = {PetMicrochipID, VaccinationStatus} Which is not a superkey thus not in BCNF, thus we decompose.

VaccinationDetails(PetMicrochipID, VaccinationStatus)
MedicalRecordDecomposed(PetMicrochipID, RecordID, VaccinationStatus, InsurancePolicyNumber, HealthCondition, VetNotes)

These are both in BCNF, thus they are our final tables.

VaccinationDetails(PetMicrochipID: INTEGER(15), VaccinationStatus: CHAR(1))
MedicalRecordDecomposed(PetMicrochipID: INTEGER(15), InsurancePolicyNumber: INTEGER(20), RecordID: INTEGER(20), HealthCondition:VARCHAR , VetNotes: VARCHAR)

AdoptionCenter:

CenterLicenseNumber -> CenterName, Address, AnimalCapacity

This tables is already in BCNF, thus remains:

AdoptionCenter(CenterLicenseNumber: INTEGER(10), CenterName: VARCHAR, Address: VARCHAR, AnimalCapacity: INTEGER)

Species:

SpeciesName -> HousingSpaceRequired, GroomingRoutine, DietType

This tables is already in BCNF, thus remains:

Species(SpeciesName: VARCHAR, HousingSpaceRequired: VARCHAR, GroomingRoutine: VARCHAR, DietType: VARCHAR)

Veterinarian:

VetLicenseNumber -> Name, ClinicName, ContactNumber, EmailAddress

This tables is already in BCNF, thus remains:

Veterinarian(VetLicenseNumber: INTEGER(10), Name: VARCHAR NOT NULL, ClinicName: VARCHAR, ContactNumber: INTEGER, EmailAddress: VARCHAR)

Employee:

SIN -> FirstName, LastName

This table is already in BCNF, thus remains

Employee(SIN: INTEGER(9), FirstName: VARCHAR, LastName: VARCHAR)

OwnerAdoptsPetFromAdoptionCenter:

PetMicrochipID, AdoptionDate-> ClientID
PetMicrochipID, ClientID -> AdoptionDate

PetMicrochipID, ClientID+ = {PetMicrochipID, ClientID, AdoptionDate}, thus it is a superkey and already is in BCNF. No decompositions are needed.

The table remains:

OwnerAdoptsPetFromAdoptionCenter(**PetMicrochipID:** CHAR(15), AdoptionDate: CHAR(8), **ClientID**: INTEGER(10))

OwnerGivesUpPetToAdoptionCenter:
PetMicrochipID, Date -> ClientID, CenterLicenseNumber, Reason
PetMicrochipID, ClientID -> Date, Reason, CenterLicenseNumber
PetMicrochipID, CenterLicenseNumber -> Date, ClientID, Reason

This table is in BCNF, as all FDs are superkeys, thus remains:

OwnerGivesUpPetToAdoptionCenter(**PetMicrochipID:** INTEGER(15), **ClientID**: INTEGER(10), Date: CHAR(8), Reason: VARCHAR)

DonorDonatesToAdoptionCenter:
ClientID,CenterLicenseNumber, Date -> Amount

This table is already in BCNF, thus remains:

DonorDonatesToAdoptionCenter(**ClientID**: INTEGER(10), CenterLicenseNumber: INTEGER(10), Date: CHAR(8), Amount: INTEGER)

EmployeeCaresForPet:
This table has no non-trivial functional dependencies, and thus is in BCNF. It remains:

EmployeeCaresForPet(**PetMicrochipID:** INTEGER(15), **SIN:** INTEGER(9))

EmployeeWorksAtAdoptionCenter:
SIN, CenterLicenseNumber -> Role, Salary
CenterLicenseNumber, Role -> Salary

CenterLicenseNumber, Role+ = {CenterLicenseNumber, Role, Salary} which is not a superkey and thus not in BCNF. We must decompose:

CenterRoleSalary(CenterLicenseNumber, Role, Salary)
EmployeeWorksAtAdoptionCenterDecomposed(SIN, CenterLicenseNumber, Role)

These are in BCNF, thus our final tables are:

CenterRoleSalary(**CenterLicenseNumber**: INTEGER(10), Role: VARCHAR, Salary: INTEGER)

EmployeeWorksAtAdoptionCenterDecomposed(**SIN:** INTEGER(9), **CenterLicenseNumber:** INTEGER(10), **Role**: VARCHAR)

VeterinarianSpecializesInSpecies:

This table has no non-trivial functional dependencies, and thus is in BCNF. It remains:

VeterinarianSpecializesInSpecies(**VetLicenseNumber:** INTEGER(10), **SpeciesName**: VARCHAR)

VeterinarianTreatsMedicalRecord:

PetMicrochipID, RecordID, VetLicenseNumber -> , Date

Our table is already in BCNF and thus remains

VeterinarianTreatsMedicalRecord(**PetMicrochipID**: INTEGER(15), **RecordID:** INTEGER(20), **VetLicenseNumber:** INTEGER(10), Date: CHAR(8))

PetHasAMedicalRecordAndInsurancePolicy:

This table has no non-trivial functional dependencies, and thus is in BCNF. It remains:

PetHasAMedicalRecordAndInsurancePolicy(**PetMicrochipID**: INTEGER(15), **InsurancePolicyNumber:** INTEGER(20), **RecordID:** INTEGER(20))

7. The SQL DDL statements required to create all the tables from item #6. The statements should use the appropriate foreign keys, primary keys, UNIQUE constraints, etc. Unless you know that you will always have exactly x characters for a given character, it is better to use the VARCHAR data type as opposed to a CHAR(Y). For example, UBC courses always use four characters to represent which department offers a course. In that case, you will want to use CHAR(4) for the department attribute in your SQL DDL statement. If you are trying to represent the name of a UBC course, you will want to use VARCHAR as the number of characters in a course name can vary greatly.

```
CREATE TABLE Client (
    ClientID INTEGER(10),
    FirstName VARCHAR NOT NULL,
    LastName VARCHAR,
    Address VARCHAR,
    ContactNumber INTEGER,
    PRIMARY KEY (ClientID));
```

```
CREATE TABLE Pet (
    PetMicrochipID INTEGER(15),
    Name VARCHAR,
    Age INTEGER,
    Breed VARCHAR NOT NULL,
    Gender CHAR(1) NOT NULL,
    PRIMARY KEY (PetMicrochipID));

CREATE TABLE Owner (
    ClientID INTEGER,
    PRIMARY KEY (ClientID),
    FOREIGN KEY (ClientID) REFERENCES Client(ClientID));

CREATE TABLE Donor (
    ClientID INTEGER,
    PRIMARY KEY (ClientID),
    FOREIGN KEY (ClientID) REFERENCES Client(ClientID));
```

InsuranceDates(<u>InsuranceStartDate :</u> CHAR(8), InsuranceExpiration: CHAR(8))

```
CREATE TABLE InsuranceDates (
        InsuranceStartDate CHAR(8) PRIMARY KEY,
        InsuranceExpiration CHAR(8));
```

VaccinationDetails(<u>PetMicrochipID</u>: INTEGER(15), VaccinationStatus: CHAR(1))
MedicalRecordDecomposed(<u>PetMicrochipID</u>: INTEGER(15), <u>InsurancePolicyNumber</u>: INTEGER(20), <u>RecordID</u>: INTEGER(20), HealthCondition:VARCHAR , VetNotes: VARCHAR)

```
CREATE TABLE MedicalRecordDecomposed (
        PetMicrochipID INTEGER(15),
        InsurancePolicyNumber INTEGER(20),
        RecordID INTEGER(20),
        HealthCondition VARCHAR,
        VetNotes VARCHAR,
        PRIMARY KEY (PetMicrochipID, InsurancePolicyNumber, RecordID),
        FOREIGN KEY (PetMicrochipID) REFERENCES Pet(PetMicrochipID),
        FOREIGN KEY (InsurancePolicyNumber) REFERENCES
        InsurancePolicy(InsurancePolicyNumber));

CREATE TABLE VaccinationDetails (
        PetMicrochipID INTEGER(15),
```

```
        VaccinationStatus CHAR(1),
        PRIMARY KEY (PetMicrochipID),
        FOREIGN KEY (PetMicrochipID) REFERENCES Pet(PetMicrochipID));
```

InsurancePolicyDecomposed(<u>InsurancePolicyNumber:</u> INTEGER(20), Level: VARCHAR, CoverageAmount: INTEGER, **InsuranceStartDate:** CHAR(8))

```
CREATE TABLE InsurancePolicyDecomposed (
        InsurancePolicyNumber INTEGER(20) PRIMARY KEY,
        Level VARCHAR,
        CoverageAmount INTEGER NOT NULL,
        InsuranceStartDate CHAR(8) NOT NULL,
        FOREIGN KEY (InsuranceStartDate) REFERENCES InsuranceDates);
```

AdoptionCenter(<u>CenterLicenseNumber:</u> INTEGER(10), CenterName: VARCHAR, Address: VARCHAR, AnimalCapacity: INTEGER)

```
CREATE TABLE AdoptionCenter (
        CenterLicenseNumber INTEGER(10) PRIMARY KEY,
        CenterName VARCHAR,
        Address VARCHAR,
        AnimalCapacity INTEGER);
```

Species(<u>SpeciesName:</u> VARCHAR, HousingSpaceRequired: VARCHAR, GroomingRoutine: VARCHAR, DietType: VARCHAR)

```
CREATE TABLE Species (
        SpeciesName VARCHAR PRIMARY KEY,
        HousingSpaceRequired VARCHAR,
        GroomingRoutine VARCHAR,
        DietType VARCHAR);
```

Veterinarian(<u>VetLicenseNumber:</u> INTEGER(10), Name: VARCHAR NOT NULL, ClinicName: VARCHAR, ContactNumber: INTEGER, EmailAddress: VARCHAR)

```
CREATE TABLE Veterinarian (
        VetLicenseNumber INTEGER(10) PRIMARY KEY,
        Name VARCHAR NOT NULL,
        ClinicName VARCHAR,
        ContactNumber INTEGER,
```

```
EmailAddress VARCHAR);
```

CenterRoleSalary(**CenterLicenseNumber**: INTEGER(10), Role: VARCHAR, Salary: INTEGER)

```
CREATE TABLE CenterRoleSalary (
    CenterLicenseNumber INTEGER(10),
    Role VARCHAR,
    Salary INTEGER,
    PRIMARY KEY (CenterLicenseNumber, Role),
    FOREIGN KEY (CenterLicenseNumber) REFERENCES AdoptionCenter);
```

Employee(SIN: INTEGER(9), FirstName: VARCHAR, LastName: VARCHAR)

```
CREATE TABLE Employee (
    SIN INTEGER(9) PRIMARY KEY,
    FirstName VARCHAR,
    LastName VARCHAR);
```

OwnerAdoptsPetFromAdoptionCenter(**PetMicrochipID**: CHAR(15), AdoptionDate: CHAR(8), **ClientID**: INTEGER(10))

```
CREATE TABLE OwnerAdoptsPetFromAdoptionCenter (
    PetMicrochipID CHAR(15),
    AdoptionDate CHAR(8),
    ClientID INTEGER(10),
    PRIMARY KEY (PetMicrochipID, AdoptionDate),
    FOREIGN KEY (PetMicrochipID) REFERENCES Pet(PetMicrochipID),
    FOREIGN KEY (ClientID) REFERENCES Client(ClientID));
```

OwnerGivesUpPetToAdoptionCenter(**PetMicrochipID:** INTEGER(15), **ClientID**: INTEGER(10), Date: CHAR(8), Reason: VARCHAR)

```
CREATE TABLE OwnerGivesUpPetToAdoptionCenter (
    PetMicrochipID INTEGER(15),
    ClientID INTEGER(10),
    Date CHAR(8),
    Reason VARCHAR,
    PRIMARY KEY (PetMicrochipID, ClientID),
```

```
        FOREIGN KEY (PetMicrochipID) REFERENCES Pet,
        FOREIGN KEY (ClientID) REFERENCES Client(ClientID));
```

DonorDonatesToAdoptionCenter(**ClientID**: INTEGER(10), CenterLicenseNumber: INTEGER(10), Date: CHAR(8), Amount: INTEGER)

```
CREATE TABLE DonorDonatesToAdoptionCenterDecomposed (
        ClientID INTEGER(10),
        CenterLicenseNumber INTEGER(10),
        Date CHAR(8),
        Amount INTEGER,
        PRIMARY KEY (ClientID, CenterLicenseNumber, Date),
        FOREIGN KEY (ClientID) REFERENCES Client(ClientID),
        FOREIGN KEY (CenterLicenseNumber) REFERENCES AdoptionCenter);
```

EmployeeCaresForPet(**PetMicrochipID:** INTEGER(15), **SIN:** INTEGER(9))

```
CREATE TABLE EmployeeCaresForPet (
        PetMicrochipID INTEGER(15),
        SIN INTEGER(9),
        PRIMARY KEY (PetMicrochipID, SIN),
        FOREIGN KEY (SIN) REFERENCES EmployeeInfo),
        FOREIGN KEY (PetMicrochipID) REFERENCES Pet);
```

EmployeeWorksAtAdoptionCenterDecomposed(**SIN:** INTEGER(9), **CenterLicenseNumber:** INTEGER(10), **Role**: VARCHAR)

```
CREATE TABLE EmployeeWorksAtAdoptionCenter (
        SIN INTEGER(9),
        CenterLicenseNumber INTEGER(10),
        Role VARCHAR
        PRIMARY KEY (SIN, CenterLicenseNumber),
        FOREIGN KEY (SIN) REFERENCES EmployeeInfo,
        FOREIGN KEY (CenterLicenseNumber) REFERENCES AdoptionCenter,
        FOREIGN KEY (Role) REFERENCES CenterRoleSalary);
```

VeterinarianSpecializesInSpecies(**VetLicenseNumber:** INTEGER(10), **SpeciesName**: VARCHAR)

```
CREATE TABLE VeterinarianSpecializesInSpecies (
    VetLicenseNumber INTEGER(10),
    SpeciesName VARCHAR,
    PRIMARY KEY (VetLicenseNumber, SpeciesName),
    FOREIGN KEY (VetLicenseNumber) REFERENCES Veterinarian,
    FOREIGN KEY (SpeciesName) REFERENCES Species);
```

VeterinarianTreatsMedicalRecord(**PetMicrochipID**: INTEGER(15), **RecordID:** INTEGER(20), **VetLicenseNumber:** INTEGER(10), Date: CHAR(8))

```
CREATE TABLE VeterinarianTreatsMedicalRecord (
    PetMicrochipID INTEGER(15),
    RecordID INTEGER(20),
    VetLicenseNumber INTEGER(10),
    PRIMARY KEY (PetMicrochipID, RecordID, VetLicenseNumber),
    FOREIGN KEY (PetMicrochipID) REFERENCES Pet,
    FOREIGN KEY (RecordID) REFERENCES MedicalRecordDecomposed
    FOREIGN KEY (VetLicenseNumber) REFERENCES Veterinarian);
```

PetHasAMedicalRecordAndInsurancePolicy(**PetMicrochipID**: INTEGER(15), **InsurancePolicyNumber:** INTEGER(20), **RecordID:** INTEGER(20))

```
CREATE TABLE PetHasAMedicalRecordAndInsurancePolicy (
    PetMicrochipID INTEGER(15),
    InsurancePolicyNumber INTEGER(20),
    RecordID INTEGER(20),
    PRIMARY KEY (PetMicrochipID, InsurancePolicyNumber, RecordID),
    FOREIGN KEY (PetMicrochipID) REFERENCES Pet,
    FOREIGN KEY (InsurancePolicyNumber) REFERENCES
InsurancePolicyDecomposed,
    FOREIGN KEY (RecordID) REFERENCES MedicalRecordDecomposed);
```

8. INSERT statements to populate each table with at least 5 tuples. You will likely want to have more than 5 tuples so that you can have meaningful queries later.

1. **Client**:

```
INSERT INTO Client (ClientID, FirstName, LastName, Address, ContactNumber) VALUES
        (0000000001, 'Alice', NULL, '5960 Student Union Blvd.', 6041234567),
        (0000000002, 'Mickey', 'Carson', '2233 Naples Dr.', 7781234567),
        (0000000003, 'Jang', 'Uk', '456 Oak Ave.', 6041334455),
        (0000000004, 'Kim', 'Jong-Im', '6088 Walter Gage Rd.', 6045555678),
        (0000000005, 'Tyson', 'Miller', '788 Yew St.', 6045545607),
        (0000000006, 'Lee', 'Johnson', '590 Joyce Blvd.', 6044561237),
        (0000000007, 'Tom', 'Holland', '213 Olivia Dr.', 7784561237),
        (0000000008, 'Robert', 'Downey Jr.', '221 Mark Ave.', 6044413355),
        (0000000009, 'Amanda', 'Pellerine', '1101 West Mall', 6045412293),
        (0000000010, 'Stacy', 'Brown', '576 Hastings St.', 6042391283);
```

2. **Pet**:

```
INSERT INTO Pet (PetMicrochipID, Name, Age, Breed, Gender) VALUES
        (123123123123123, 'Willow', 3, 'Golden Retriever', 'M'),
        (100200300405067, 'Mowgli', 8, 'Goldendoodle', 'F'),
        (100300400578563, 'Boss', 3, 'Pomeranian', 'M'),
        (236293424532003, 'Luna', 1, 'Tabby', 'F'),
        (100519632323972, 'Frenchy', 5, 'Beagle', 'M');
```

3. **Owner**:

```
INSERT INTO Owner (ClientID) VALUES
        (0000000001),
        (0000000002),
        (0000000003),
        (0000000004),
        (0000000005);
```

4. **Donor**:

```
INSERT INTO Donor (ClientID) VALUES
        (0000000006),
        (0000000007),
        (0000000008),
        (0000000009),
        (0000000010);
```

5. **InsuranceDates**:

```
INSERT INTO InsuranceDates (InsuranceStartDate, InsuranceExpiration) VALUES
```

('20230101', '20240101'),
('20230201', '20240201'),
('20230301', '20240301'),
('20230401', '20240401'),
('20230501', '20240501');

6. **InsurancePolicyDecomposed**:

INSERT INTO InsurancePolicyDecomposed (InsurancePolicyNumber, Level, CoverageAmount, InsuranceStartDate) VALUES
(0001, 'Basic', 1000, '20230101'),
(0002, 'Premium', 2500, '20230201'), (0003, 'Platinum' , 10000, '20240503'), (0004, 'Silver', 1500, '20220401'), (0005, 'Bronze', 1200, '20100301')

7. **MedicalRecordDecomposed**:

INSERT INTO MedicalRecordDecomposed (PetMicrochipID, InsurancePolicyNumber, RecordID, HealthCondition, VetNotes) VALUES
(123123123123123, 0001, 3001, 'Healthy', 'Regular check-up'),
(100200300405067, 0002, 3002, 'Allergic to chicken', 'Prescribed antihistamines'),
(100300400578563, 0003, 3003, 'Arthritis', 'Recommend physical therapy'),
(236293424532003, 0004, 3004, 'Injury on left knee', 'Surgery required'),
(100519632323972, 0005, 3005, 'Obesity', 'Diet and exercise advised');

8. **VaccinationDetails:**

INSERT INTO VaccinationDetails (PetMicrochipID, VaccinationStatus) VALUES
(123123123123123, 'Y'),
(100200300405067, 'N'),
(100300400578563, 'Y'),
(236293424532003, 'Y'),
(100519632323972, 'N');

9. **AdoptionCenter**:

INSERT INTO AdoptionCenter (CenterLicenseNumber, CenterName, Address, AnimalCapacity) VALUES
(5001, 'Happy Tails', '111 Dane Way', 50),
(5002, 'Paws & Claws', '222 Seymour St', 30),
(5003, 'Forever Home', '333 Drummond Ave', 40),
(5004, 'City of Vancouver Animal Services', '1280 Raymur Ave', 60),
(5005, 'Vancouver Pound', '800 Hamilton St', 45);

10. Species

INSERT INTO Species (SpeciesName, HousingSpaceRequired, GroomingRoutine, DietType) VALUES
('Dog', '250sqft', 'brush daily, haircut every 4 to 8 weeks', 'Kibble'),
('Cat', '300sqft', 'wash every 4 to 6 weeks', 'obligate carnivore'),
('Rabbit','64sqft', 'brush daily, don't bathe', 'hay, fresh greens, pellets'),
('Goldfish', '20 gallons/fish', 'change water every week', 'omnivorous, goldfish flakes and pellets'),
('Hamster', '5 sqft', 'self-grooming, occasional cleaning when necessary', 'fresh greens, hamster pellets');

11. **Veterinarian**:

INSERT INTO Veterinarian (VetLicenseNumber, Name, ClinicName, ContactNumber, EmailAddress) VALUES
(7001, 'Dr. Chelsea', 'PetSmart', 778 028 2382, 'chelsea@petsmart.com'),
(2923, 'Dr. Ten', 'PetSmart', 778 208 2309, 'ten@petsmart.com'),
(4584, 'Dr. Isha', 'PetCo', 604 028 2039, 'isha@petco.com'),
(3949, 'Dr. Daniel', 'SPCA Vancouver', 408 304 3084, 'daniel@spcavan.ca'),
(8732, 'Dr. Arisu', 'RAPS Vancouver', 778 238 2938, 'arisu@raps.ca');

12. **CenterRoleSalary**:

INSERT INTO CenterRoleSalary (CenterLicenseNumber, Role, Salary) VALUES
(5001, 'Manager', 60000),
(5001, 'Caretaker', 50000),
(5005, 'Receptionist', 45000),
(5003, 'Volunteer', 0),
(5004, 'Adoption Counselors', 50000);

13. **Employee**

INSERT INTO Employee (SIN, FirstName, LastName) VALUES
(123456789, 'John', 'Doe'),
(987654321, 'Bob', 'Smith'),
(121212121, 'Freddy', 'Mercury'),
(567567567, 'Samuel', 'Jackson'),
(890890890, 'Tom', 'Hanks');

14. OwnerAdoptsPetFromAdoptionCenter:

INSERT INTO OwnerAdoptsPetFromAdoptionCenter (PetMicrochipID, AdoptionDate, ClientID) VALUES
(123123123123123, '20230101', 0001),
(100200300405067, '20230602', 0002),
(100300400578563, '20230603', 0003),
(236293424532003, '20230604', 0004),
(100519632323972, '20230605', 0005);

15. **OwnerGivesUpPetToAdoptionCenter**

INSERT INTO OwnerGivesUpPetToAdoptionCenter (PetMicrochipID, ClientID, Date, Reason) VALUES
(123123123123123, 0001, '20240101', 'Unfriendly with our children'),
(100200300405067, 0002, ‘20250209’, ‘Moving to new country’),
(100300400578563, 0003, ‘20230903’, ‘Unaffordable’),
(236293424532003, 0004, ‘20250203’, NULL),
(100519632323972, 0005, '20230705', 'Incompatible personalities');

16. DonorDonatesToAdoptionCenter

INSERT INTO DonorDonatesToAdoptionCenter (ClientID, CenterLicenseNumber, Date, Amount) VALUES
(0000000007, 5001, ‘20250101’, 100),
(0000000007, 5001, ‘20240501’, 1000),
(0000000008, 5004, ‘20200306’, 20),
(0000000009, 5004, ‘20100909’, 50),
(0000000010, 5002, ‘19990602’, 25000);

17. EmployeeCaresForPet

INSERT INTO EmployeeCaresForPet (PetMicrochipID, SIN) VALUES
(123123123123123, 123456789),
(100200300405067, 987654321),
(100300400578563, 121212121),
(236293424532003, 567567567),
(100519632323972, 890890890);

18. EmployeeWorksAtAdoptionCenterDecomposed

INSERT INTO EmployeeWorksAtAdoptionCenterDecomposed (SIN, CenterLicenseNumber, Role) VALUES
(123456789, 5001, 'Animal Care Attendant'),
(987654321, 5002, 'Manager'),

(121212121, 5001, 'Customer Service Representative),
(567567567, 5004, 'Kennel Attendant'),
(890890890, 5005, 'Animal Protection Officer');

19. **VeterinarianSpecializesInSpecies**:

INSERT INTO VeterinarianSpecializesInSpecies (VetLicenseNumber, SpeciesName) VALUES
(7001, 'Canine'),
(2923, 'Canine'),
(4584, 'Rodent'),
(3949, 'Reptile'),
(8732, 'Feline');

20. VeterinarianTreatsMedicalRecord:

INSERT INTO VeterinarianTreatsMedicalRecord (PetMicrochipID, RecordID, VetLicenseNumber) VALUES
(123123123123123, 3001, 7001),
(100200300405067, 3002, 2923),
(100300400578563, 3003, 7001),
(236293424532003, 3004, 2923),
(100519632323972, 3005, 8732);

21. **PetHasAMedicalRecordAndInsurancePolicy**:

INSERT INTO PetHasAMedicalRecordAndInsurancePolicy (PetMicrochipID, InsurancePolicyNumber, RecordID) VALUES
(123123123123123, 0001, 3001),
(100200300405067, 0002, 3002),
(100300400578563, 0003, 3003),
(236293424532003, 0004, 3004),
(100519632323972, 0005, 3005);

9. An explicit acknowledgment about your use of AI tools in this assignment. Specifically, we are looking for a clear yes/no about whether you have used one or more AI tools. If yes, we want to know which tool(s) you have used and what prompt(s) you have given the tool.

Yes.

Prompts:

1. Is INT and INTEGER the same thing in SQL?